



Modelling Financial Markets

Chapter 18: Systemic Risk and Networks



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BNP: BNP Paribas SA (Euronext Paris ticker; Banque Nationale de Paris)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ ECB: European Central Bank
- ▣ ES: Expected Shortfall
- ▣ EU: European Union
- ▣ FDIC: Federal Deposit Insurance Corporation
- ▣ FDR: False Discovery Rate
- ▣ FRM: Financial Risk Meter
- ▣ GACV: Generalized Approximate Cross-Validation
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HSBC: HSBC Holdings plc (originally the Hongkong and Shanghai Banking Corporation)
- ▣ ING: ING Groep (Internationale Nederlanden Groep) (the Dutch banking group ING)
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ LRMEs: Long-Run Marginal Expected Shortfall
- ▣ MES: Marginal Expected Shortfall
- ▣ OLS: Ordinary Least Squares
- ▣ SG: Société Générale (the French banking group Société Générale)
- ▣ SRISK: Systemic RISK (capital shortfall measure)
- ▣ SVB: Silicon Valley Bank
- ▣ TENET: Tail-Event driven NETwork risk
- ▣ US: United States
- ▣ VAR: Vector AutoRegression
- ▣ VIX: CBOE Volatility Index

Outline

Motivation

Exposure: MES and SRISK

Contribution: CoVaR

Connectedness: Spillover Networks

Tail Networks: the Financial Risk Meter

Climate Risk

Stress Testing

AI for Scientific Discovery

Summary

References

What Is Systemic Risk?

From one bank to the system

- ▣ **Systemic risk**: the risk that the distress of one or several institutions impairs the financial system as a whole, and through it the real economy
- ▣ It is an **externality**: a bank does not pay for the losses its failure imposes on others
- ▣ Chapter 7 measured the risk **of** a portfolio; this chapter measures the risk a bank **adds to** or **takes from** the system

What we do in this chapter

- ▣ Measure the exposure of banks to market crashes: MES and SRISK
- ▣ Measure the contribution of a bank to system losses: CoVaR and ΔCoVaR
- ▣ Map the network of volatility spillovers (Diebold–Yilmaz) and tail links (the Financial Risk Meter)
- ▣ Design stress tests: historical, hypothetical and reverse; climate risk

Northern Rock, September 2007: a Bank Run



Depositors queue outside a Northern Rock branch, Brighton,
14 September 2007

Photo: Dominic Alves (2007); CC BY 2.0; Wikimedia Commons

- A British mortgage lender hit by the 2007 freeze of funding markets that followed the US subprime losses; depositors queued to withdraw their savings
- Channels of contagion
 - ▶ **Funding runs:** depositors and lenders withdraw from any bank that looks similar
 - ▶ **Common exposures and fire sales:** forced selling lowers prices for every holder
 - ▶ **Counterparty links:** the loss of one bank is an asset of another

Silicon Valley Bank, March 2023: a Digital Run



Headquarters of Silicon Valley Bank, Santa Clara, 13 March 2023

Photo: Minh Nguyen (2023); CC BY-SA 4.0; Wikimedia Commons

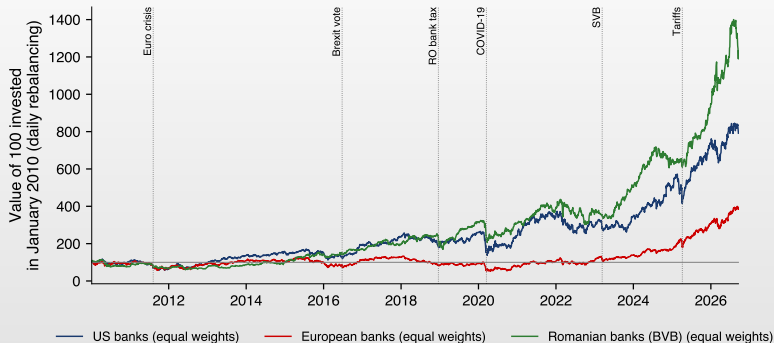
- The FDIC took over Silicon Valley Bank on 10 March 2023 and created a bridge bank for its insured deposits [FDIC \(2023\)](#)
- Two days later US authorities decided to protect **all** depositors of SVB and Signature Bank, insured and uninsured [Federal Reserve Board \(2023\)](#)
 - ▶ The decision rests on a **systemic** argument: to reduce stress across the financial system
- Bank shares fell worldwide in the following days; Section 7 measures the loss of a bank portfolio in that window

Data Used in This Chapter

Group	Series	Market index
US banks	JPMorgan, Bank of America, Citigroup, Goldman Sachs, Morgan Stanley, Wells Fargo	S&P 500
European banks	Deutsche Bank, BNP Paribas, Santander, ING, HSBC	Euro Stoxx 50
Romanian banks	Banca Transilvania, BRD-Groupe Société Générale	BET
Volatility	the VIX index (implied volatility of the S&P 500)	

- Daily prices from January 2010 to 18 September 2026, adjusted for dividends and splits; 3 960 common trading days
- Prices are joined on common days first, then daily log returns $r_t = 100(\ln P_t - \ln P_{t-1})$, in %
- Romanian banks: days without trades removed; historical stress scenarios for the international banks start in 2007

Three Banking Systems, 2010–2026



- Equal-weighted portfolios of the banks of each region, rebalanced daily; vertical lines mark the episodes used in this chapter

Reading the Three Banking Systems

- ▣ 100 invested in January 2010 became 791 in US banks, 384 in European banks and 1233 in Romanian banks
 - ▶ Largest peak-to-trough fall: US 51% (2011-11-23), Europe 61% (2020-04-21), Romania 47% (2012-06-06)
- ▣ Annualised volatility of the portfolios: 29%, 30% and 23%, against 18% (S&P 500), 21% (Euro Stoxx 50) and 18% (BET)
- ▣ The falls are **simultaneous**: the euro crisis, COVID-19 and the 2023 bank failures hit all regions in the same weeks
- ▣ Romanian banks also have their own shock: the bank-tax announcement of December 2018

Conventions and a Map of Systemic Risk Measures

- Return X , loss $L = -X$; the level α is a **tail probability**: VaR 1%, CoVaR 1%, MES 5% (Chapter 7)
 - ▶ $\text{VaR}_\alpha(X) = -q_\alpha(X)$, with q_α the α -quantile; VaR, CoVaR and MES are positive numbers when they are losses
- **System** return X^{sys} : the equal-weighted portfolio of the other banks of the same region
- Three families of measures Billio, Getmansky, Lo & Pelizzon (2012)
 - ▶ **Exposure**: how much does the bank lose when the market or the system crashes? MES, SRISK, exposure CoVaR
 - ▶ **Contribution**: how much worse is the system when the bank is in distress? ΔCoVaR
 - ▶ **Connectedness**: through which links do shocks travel? Spillover networks, tail networks
- Textbooks: Franke, Härdle & Hafner (2019), Härdle, Chen & Overbeck (2017)

Recap: Motivation

- ▣ Systemic risk is about the system, not the single bank: it lives in the joint tail of many institutions
- ▣ Runs, common exposures, fire sales and counterparty links turn one failure into many
- ▣ Bank portfolios fall together in crises and are more volatile than the market
- ▣ We need measures of exposure, contribution and connectedness, each with its uncertainty

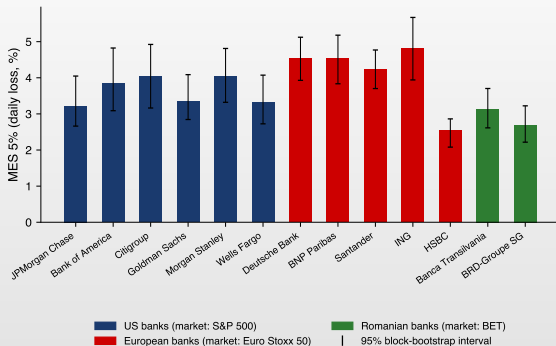
Marginal Expected Shortfall

- **MES** (Marginal Expected Shortfall) of bank i at level α Acharya, Pedersen, Philippon & Richardson (2017)
 - ▶ $\text{MES}_{i,\alpha} = -\mathbb{E}[X_i \mid X_m \leq q_\alpha(X_m)]$
 - ▶ The average loss of the bank on the worst α share of days for the market m
- Why “marginal”: for a portfolio with weights w_i , $\text{ES}_\alpha(X_p) = \sum_i w_i \text{MES}_{i,\alpha}$
 - ▶ MES is the contribution of bank i to the Expected Shortfall of the market (Euler allocation, Chapter 7)
- Historical estimate: average $-X_i$ over the days with $X_m \leq \hat{q}_\alpha(X_m)$; with $\alpha = 5\%$ and 3 960 days, about 200 days
- Market m : S&P 500 for US banks, Euro Stoxx 50 for European banks, BET for Romanian banks

Worked Example: MES under the Normal Distribution

- If (X_i, X_m) are jointly Normal with zero means, $\mathbb{E}[X_i | X_m] = \beta_i X_m$ with $\beta_i = \text{Cov}(X_i, X_m) / \text{Var}(X_m)$
 - ▶ $\text{MES}_{i,\alpha} = \beta_i \text{ES}_\alpha(X_m) = \beta_i \sigma_m \frac{\varphi(z_\alpha)}{\alpha}$, with $z_\alpha = \Phi^{-1}(\alpha)$
- Example: $\beta_i = 1.2$, $\sigma_m = 1.5\%$ per day, $\alpha = 5\%$
 - ▶ $z_{5\%} = -1.645$, $\varphi(z_{5\%})/0.05 = 2.063$, $\text{MES}_{i,5\%} = 1.2 \times 1.5 \times 2.063 = 3.71\%$
- Under the Normal distribution MES is just beta times a constant; with real data the tail relation can be stronger than beta suggests

MES 5% of Thirteen Banks



- Average daily loss of each bank on the worst 5% of days for its market index, 2010–2026; 95% intervals from a moving-block bootstrap (blocks of 20 days)

Reading the MES

- ▣ US banks: from 3.22% (JPMorgan) to 4.04% (Morgan Stanley); European banks up to 4.81% (ING)
 - ▶ On the same days the market lost on average 2.72% (S&P 500) and 3.16% (Euro Stoxx 50): banks lose more than the market
- ▣ HSBC is the exception in Europe (2.54%, below the index): a London listing with Asian business
- ▣ Banca Transilvania 3.14% and BRD 2.69% on BET crash days; BET itself lost 2.70%
- ▣ The intervals overlap within each region: the data rank the regions more clearly than the banks

From MES to a Crisis: LRMES

- A crisis is a fall of the market by 40% over six months; the **LRMES** (Long-Run MES) is the expected fall of the bank's equity in such a crisis
 - ▶ Approximation from daily data [Acharya, Engle & Richardson \(2012\)](#): $\text{LRMES}_i \approx 1 - \exp(-18 \times \text{MES}_i^{2\%})$
 - ▶ $\text{MES}_i^{2\%}$: average daily loss of the bank on days when the market falls by more than 2% (in decimals)
- Example: $\text{MES}^{2\%} = 3\%$ gives $\text{LRMES} = 1 - e^{-0.54} = 41.7\%$
- Our banks: LRMES from 37% (HSBC) to 58% (ING); JPMorgan 48%, Banca Transilvania 52%
 - ▶ The market fell by more than 2% on 138 days in the US, 196 in the euro area and 111 on the BVB

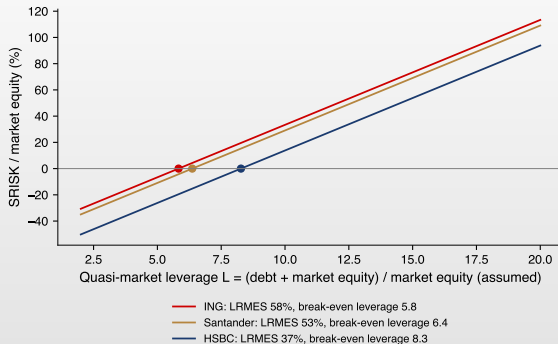
SRISK: the Capital Shortfall in a Crisis

- A bank must keep equity of at least a share k of its assets (debt D plus equity W); $k = 8\%$ [Brownlees & Engle \(2017\)](#)
 - ▶ $SRISK_i = k D_i - (1 - k)(1 - LRMES_i) W_i$
 - ▶ The capital the bank would need to raise after a crisis in which its equity falls by LRMES and its debt does not change
- Positive SRISK = shortfall; the SRISK of the system is the sum of the positive values; [V-Lab, New York University Stern School of Business](#) publishes it for large banks with their balance sheets
- Example: $D = 900$, $W = 100$, $LRMES = 41.7\%$
 - ▶ $SRISK = 0.08 \times 900 - 0.92 \times (1 - 41.7\%) \times 100 = 72 - 53.6 = 18.4$
- SRISK combines a market measure (LRMES) with a balance-sheet measure (leverage)

SRISK per Unit of Equity and the Break-Even Leverage

- Divide by W and write the quasi-market leverage $L = (D + W)/W$
 - ▶ $\frac{\text{SRISK}}{W} = k(L - 1) - (1 - k)(1 - \text{LRMES})$
- SRISK is positive once leverage exceeds the **break-even leverage**
 - ▶ $L^* = 1 + \frac{(1 - k)(1 - \text{LRMES})}{k}$
 - ▶ Example above: $L^* = 7.7$, while the bank has $L = 10$: a shortfall
- Our data contain prices, not balance sheets: we report L^* from the estimated LRMES and treat leverage as an **assumption**
- A higher LRMES lowers L^* : the bank can afford less debt before a crisis creates a shortfall

SRISK as a Function of Leverage



- Banks with the highest, median and lowest LRMES; leverage on the horizontal axis is assumed, not observed; dots mark L^*

Reading the Break-Even Leverage

- With $k = 8\%$: $L^* = 5.8$ for ING, 6.4 for Santander, 8.3 for HSBC
 - ▶ US: JPMorgan 7.0, Morgan Stanley 6.0; Romania: Banca Transilvania 6.5, BRD 7.1
- A bank whose debt is more than about 5–7 times its market equity would need new capital in a 40% market crash
- With $k = 5.5\%$ the thresholds rise (ING 8.2, HSBC 11.9): the prudential ratio is a policy choice, and SRISK depends on it
- Limits: LRMES is an extrapolation from daily data, and market equity itself falls in a crisis

Recap: MES and SRISK

- MES is the bank's average loss on the market's worst days; it adds up to the market's ES
- LRMES extrapolates MES to a six-month crisis; SRISK turns it into the capital shortfall
- European banks have the largest MES in our sample; bank and region rankings carry wide intervals
- SRISK needs balance-sheet data; without it we can report only the break-even leverage

CoVaR: the VaR of the System Given a Bank in Distress

- **CoVaR** of the system conditional on bank i at its VaR [Adrian & Brunnermeier \(2016\)](#)
 - ▶ $P\left(X^{sys} \leq -\text{CoVaR}_{\alpha}^{sys|i} \mid X_i = -\text{VaR}_{i,\alpha}\right) = \alpha$
 - ▶ The loss of the system that is exceeded with probability α on a day when bank i is at its own VaR
- **ΔCoVaR** : the change from the median state of the bank to its distress state
 - ▶ $\Delta\text{CoVaR}_{\alpha}^{sys|i} = \text{CoVaR}_{\alpha}^{sys|X_i=q_{\alpha}} - \text{CoVaR}_{\alpha}^{sys|X_i=q_{50\%}}$
- ΔCoVaR measures the **contribution** of bank i to the tail risk of the system
- Reversing the roles gives the **exposure CoVaR**: the VaR of bank i when the system is in distress

Quantile Regression

- The α -quantile of Y given x is linear: $q_\alpha(Y | x) = a_\alpha + b_\alpha x$ [Koenker & Bassett \(1978\)](#)
 - ▶ $(\hat{a}_\alpha, \hat{b}_\alpha) = \arg \min_{a,b} \sum_t \rho_\alpha(y_t - a - bx_t)$, $\rho_\alpha(u) = u(\alpha - 1\{u < 0\})$
 - ▶ ρ_α is the **check function**: it weights negative residuals by $1 - \alpha$ and positive ones by α
- With $\alpha = 50\%$ it is the median regression (least absolute deviations)
- The slope b_α can change with α : the bank matters more for the system in the tail than in the centre
 - ▶ Least squares gives only one slope, for the mean

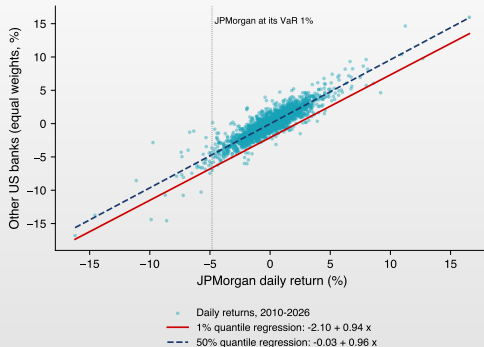
Estimating CoVaR in Three Steps

- ▣ 1. Quantile regression of the system on the bank at level α
 - ▶ $\hat{q}_\alpha(X^{sys} | X_i) = \hat{a}_\alpha + \hat{b}_\alpha X_i$
- ▣ 2. Plug in the bank's own quantiles: $\hat{q}_\alpha(X_i)$ (distress) and $\hat{q}_{50\%}(X_i)$ (median)
 - ▶ $\text{CoVaR}_\alpha^{sys|i} = - \left(\hat{a}_\alpha + \hat{b}_\alpha \hat{q}_\alpha(X_i) \right)$
- ▣ 3. Take the difference
 - ▶ $\Delta \text{CoVaR}_\alpha^{sys|i} = \hat{b}_\alpha (\hat{q}_{50\%}(X_i) - \hat{q}_\alpha(X_i))$
 - ▶ Only the slope matters: the intercept cancels
- ▣ Uncertainty: moving-block bootstrap of the days (blocks of 20 days keep volatility clusters together), 200 resamples

Worked Example: CoVaR under the Normal Distribution

- (X_i, X^{sys}) jointly Normal, zero means, standard deviations σ_i , σ_{sys} , correlation ρ
 - ▶ Given $X_i = x$: $X^{sys} \sim N\left(\rho \frac{\sigma_{sys}}{\sigma_i} x, \sigma_{sys}^2(1 - \rho^2)\right)$
 - ▶ $\text{CoVaR}_\alpha = -\sigma_{sys} z_\alpha \left(\rho + \sqrt{1 - \rho^2}\right)$, $\Delta \text{CoVaR}_\alpha = -\rho \sigma_{sys} z_\alpha$
- Example: $\sigma_{sys} = 1.5\%$, $\rho = 0.6$, $\alpha = 1\%$, $z_{1\%} = -2.326$
 - ▶ $\text{VaR}_{1\%}^{sys} = 3.49\%$, $\text{CoVaR}_{1\%} = 4.89\%$, $\Delta \text{CoVaR}_{1\%} = 0.6 \times 1.5 \times 2.326 = 2.09\%$
- ΔCoVaR does not depend on σ_i : a bank with a large VaR but a low correlation contributes little

Quantile Regression: JPMorgan and the Other US Banks

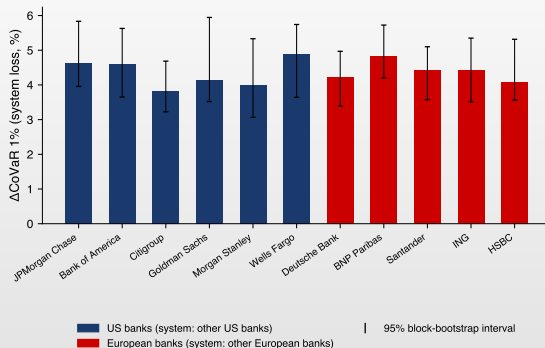


- Daily returns 2010–2026 with the 1% and 50% quantile regression lines; the dotted line marks JPMorgan at its VaR 1%

Reading the JPMorgan Example

- ▣ 1% line: $\hat{q}_{1\%} = -2.10 + 0.94 X_i$; median line: $-0.03 + 0.96 X_i$
 - ▶ Similar slopes: the 1% line is shifted down by the tail of the other banks, not steeper
- ▣ JPMorgan's VaR 1% is 4.83%; on such a day the other US banks have CoVaR 1% = 6.64%
 - ▶ Unconditional VaR 1% of the system: 5.07%; with JPMorgan at its median: 2.03%
- ▣ $\Delta\text{CoVaR}_{1\%} = 4.62\%$: JPMorgan in distress adds about 4.62 percentage points to the 1% loss of the other US banks
- ▣ This is association, not causation: a common shock hits JPMorgan and the others on the same day

ΔCoVaR 1% of Eleven Banks

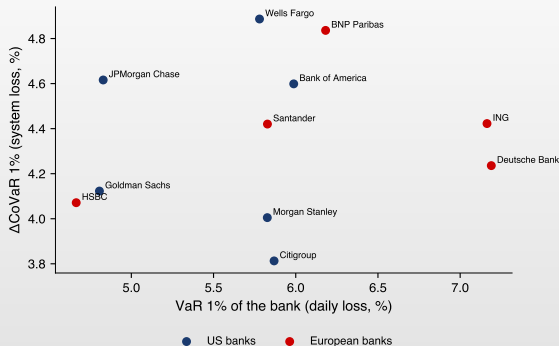


System: equal-weighted portfolio of the other banks of the same region; 95% moving-block bootstrap intervals

Reading ΔCoVaR

- All eleven banks lie between 3.81% (Citigroup) and 4.89% (Wells Fargo)
 - ▶ Every interval overlaps with every other: e.g. Citigroup [3.23, 4.69], Wells Fargo [3.64, 5.74]
- With about 40 tail days per bank, the data cannot rank these banks by their contribution
- At $\alpha = 5\%$ the values are smaller (JPMorgan 2.62%) and the ranking changes: Spearman correlation between the 1% and 5% rankings 0.10
 - ▶ A ranking that changes with α is not a robust basis for regulation

VaR and ΔCoVaR Measure Different Things



□ Horizontal: VaR 1% of the bank; vertical: its ΔCoVaR 1%, eleven banks, 2010–2026

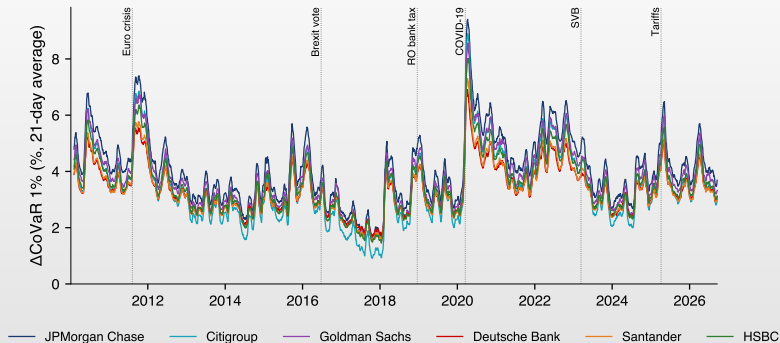
Reading VaR versus ΔCoVaR

- Spearman rank correlation between VaR and ΔCoVaR : 0.16
 - ▶ Deutsche Bank and ING have the largest VaR (7.19%, 7.16%) but only average ΔCoVaR
 - ▶ JPMorgan has one of the smallest VaR (4.83%) and one of the largest ΔCoVaR
- This is the central message of [Adrian & Brunnermeier \(2016\)](#): regulating each bank by its own VaR ignores its contribution to the system
- Caveat: with eleven banks and overlapping intervals, the correlation itself is very uncertain (Seminar 18, B4)

Time-Varying ΔCoVaR with State Variables

- State variables M_{t-1} : log VIX, the return of the regional index and of the system, all lagged one day [Adrian & Brunnermeier \(2016\)](#)
 - ▶ $q_\alpha(X_{i,t} | M_{t-1}) = a_i + c_i^\top M_{t-1}$ at α and at 50%
 - ▶ $q_\alpha(X_t^{\text{sys}} | X_{i,t}, M_{t-1}) = a + b X_{i,t} + c^\top M_{t-1}$
- Then $\Delta\text{CoVaR}_{\alpha,t} = \hat{b} (\hat{q}_{50\%,t}(X_i) - \hat{q}_{\alpha,t}(X_i))$
 - ▶ It moves with the bank's conditional VaR, which moves with the VIX
- The slope $\hat{b}$ is constant: time variation comes from the state, not from the link

$\Delta\text{CoVaR 1\%}$ over Time

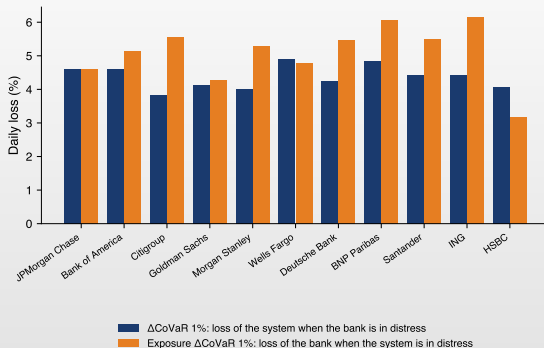


- Six banks, 21-day moving averages of the daily $\Delta\text{CoVaR 1\%}$ estimated with lagged state variables

Reading ΔCoVaR over Time

- ▣ JPMorgan: 2.16% on average in 2017, 9.06% in March 2020; peak 10.99% on 2020-03-18
 - ▶ Deutsche Bank: 2.04% in 2017 and 6.65% in March 2020
- ▣ The six lines move together: the common state variables drive them
- ▣ Market-based ΔCoVaR is low in calm periods, exactly when risk may be building up in balance sheets, which market data do not see
- ▣ [Adrian & Brunnermeier \(2016\)](#) therefore propose a forward ΔCoVaR , predicted from leverage, size and maturity mismatch

Contribution versus Exposure



- Blue: $\Delta\text{CoVaR 1\%}$ (loss of the system when the bank is in distress); orange: exposure $\Delta\text{CoVaR 1\%}$ (loss of the bank when the system is in distress)

Reading Contribution versus Exposure

- The direction matters: the two numbers differ for most banks
 - ▶ ING: contribution 4.42%, exposure 6.14%; BNP Paribas: 4.84% and 6.07%
 - ▶ HSBC: contribution 4.07%, exposure only 3.17%
- Exposure is larger than contribution for most European banks: they suffer more from a system crisis than they cause
- Exposure matters for the bank's own capital; contribution matters for the regulator of the system

Recap: CoVaR

- CoVaR is the VaR of the system given a bank in distress; ΔCoVaR is the bank's contribution
- Estimated by quantile regression; only the tail slope $\hat{b}_\alpha$ and the bank's quantiles matter
- VaR and ΔCoVaR rank banks differently; bootstrap intervals are wide and overlap
- With state variables, ΔCoVaR follows market volatility: it rises in crises and says little in calm times

From Pairs to Networks

- ▣ CoVaR looks at one bank and one system; a **network** looks at all pairs at once
 - ▶ Nodes: banks; directed edges: how much of the future uncertainty of bank i comes from shocks to bank j
- ▣ Granger-causality networks of banks, insurers, hedge funds and brokers Billio, Getmansky, Lo & Pelizzon (2012)
- ▣ Variance-decomposition networks Diebold & Yilmaz (2009), Diebold & Yilmaz (2012), Diebold & Yilmaz (2014): the approach of this section
- ▣ Tail networks from quantile regressions: TENET Härdle, Wang & Yu (2016)

The Forecast Error Variance Decomposition

- Vector autoregression (VAR) of order p for N series y_t : $y_t = c + \sum_{l=1}^p A_l y_{t-l} + \varepsilon_t$, $\text{Cov}(\varepsilon_t) = \Sigma$
 - ▶ Moving-average form: $y_t = \mu + \sum_{h=0} \Phi_h \varepsilon_{t-h}$, $\Phi_0 = I$, $\Phi_h = \sum_{l=1}^p A_l \Phi_{h-l}$
- The H -day forecast error of y_i is a sum of shocks to all series: its variance can be split by source
- Orthogonal (Cholesky) shocks depend on the ordering of the series; the **generalized** decomposition does not [Koop, Pesaran & Potter \(1996\)](#), [Pesaran & Shin \(1998\)](#)
 - ▶ $\theta_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i^\top \Phi_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} e_i^\top \Phi_h \Sigma \Phi_h^\top e_i}$, normalised: $\tilde{\theta}_{ij} = \theta_{ij} / \sum_j \theta_{ij}$

The Connectedness Table

- $\tilde{\theta}_{ij}$ (in %): the share of the H -day forecast error variance of i due to shocks in j ; each row sums to 100 [Diebold & Yilmaz \(2014\)](#)
 - ▶ **From others:** $F_i = \sum_{j \neq i} \tilde{\theta}_{ij}$; **to others:** $T_j = \sum_{i \neq j} \tilde{\theta}_{ij}$
 - ▶ **Net:** $T_i - F_i$ (positive = net transmitter); **pairwise net:** $\tilde{\theta}_{ji} - \tilde{\theta}_{ij}$
- **Total connectedness:** $C = \frac{1}{N} \sum_{i \neq j} \tilde{\theta}_{ij}$, the average share of variance that comes from others
 - ▶ Example with three banks: rows (70, 20, 10), (25, 60, 15), (5, 10, 85) give $C = (30 + 40 + 15)/3 = 28.3\%$
 - ▶ Bank 3 is a net transmitter: $T_3 = 25 > F_3 = 15$

Worked Example: from the Table to the Network

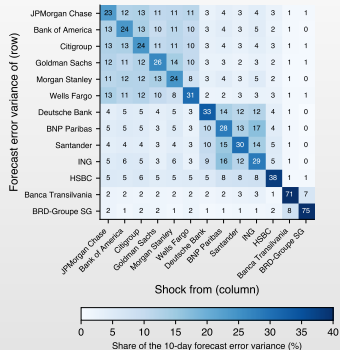
Row: variance of	1	2	3	From
Bank 1	70	20	10	30
Bank 2	25	60	15	40
Bank 3	5	10	85	15
To	30	30	25	$C = 28.3$
Net	0	-10	+10	

- Pairwise net $\tilde{\theta}_{ji} - \tilde{\theta}_{ij}$: what j sends to i minus what it receives
 - ▶ 1 and 2: $25 - 20 = +5$, an arrow $1 \rightarrow 2$
 - ▶ 3 and 1: $10 - 5 = +5$, an arrow $3 \rightarrow 1$; 3 and 2: $15 - 10 = +5$, an arrow $3 \rightarrow 2$
- Bank 3 receives little but sends to both others: the net transmitter, although its total links are the smallest
- Bank 1 is balanced ($T = F$) but still a hub: it passes shocks from 3 on to 2

What We Connect: Daily Range Volatility

- **Parkinson volatility** from the daily high H_t and low L_t [Parkinson \(1980\)](#)
 - ▶ $\hat{\sigma}_t^2 = \frac{(\ln H_t - \ln L_t)^2}{4 \ln 2}$, annualised $100\sqrt{252 \hat{\sigma}_t^2}$, in logs
 - ▶ Unbiased for the variance of a Brownian motion without drift; more efficient than the squared return
- Volatility spills over more than returns and is more forecastable: a natural object for a VAR
- Our model: VAR on the log volatility of 13 banks, order chosen by the Bayesian information criterion ($p = 1$), horizon $H = 10$ days
 - ▶ Time zones: US banks close after European ones; the VAR lags capture the next-day transmission

The Connectedness Table of Thirteen Banks

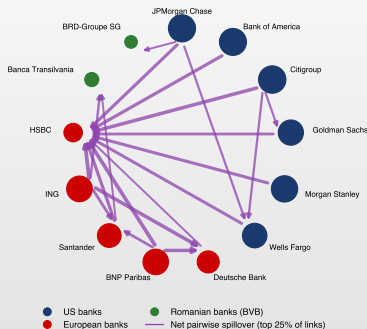


□ Row i , column j : share (%) of the 10-day forecast error variance of bank i due to shocks in bank j , full sample

Reading the Connectedness Table

- Total connectedness $C = 64.8\%$: on average two thirds of a bank's volatility risk comes from the other banks
 - ▶ On returns instead of volatility: $C = 75.4\%$
- Two blocks: US banks receive 83.0% from each other, European banks 72.2% from each other
 - ▶ Across the Atlantic: a European bank receives 26.6% from US banks, a US bank 15.7% from European ones
- Romanian banks are largely their own: own share 71.2% (Banca Transilvania) and 74.6% (BRD); they receive 11.1% from US and 8.7% from European banks

The Net Spillover Network



- Arrow from j to i when j transmits net to i (largest 25% of pairwise net spillovers); node size grows with the spillover to others

Reading the Network

- Net transmitters: Citigroup (+15.6), JPMorgan (+12.6), Bank of America (+12.4)
 - ▶ To others: Citigroup 91.4%, JPMorgan 89.4% (the sum of a column can exceed 100)
- Net receivers: HSBC (−25.8), BRD (−13.8), Banca Transilvania (−12.0), Deutsche Bank (−11.3)
 - ▶ HSBC receives from almost every bank: it trades in London, between Asia and New York
- Within Europe, BNP Paribas and ING transmit to Deutsche Bank and Santander
- A full-sample network averages quiet and crisis years: the next chart lets it move

Connectedness over Time



- Top: total connectedness on moving 250-day windows ($\text{VAR}(2)$, $H = 10$, every 5 days); bottom: net spillover summed by region

Reading Connectedness over Time

- The total index ranges from 47.4% (2024-07-24) to 79.3% (2020-08-27); mean 62.0%
 - ▶ COVID-19: from 61.8% in 2019 to 78.4% in spring 2020; it jumps within days and decays over a year
 - ▶ After the 2023 bank failures 66.0%; the calm year 2024 53.5%
- US banks are net transmitters in 90% of the windows (average net +48.5)
 - ▶ European banks were net transmitters during the euro crisis of 2011–2012 and mostly net receivers afterwards
- Romanian banks are net receivers on average (−24.1)

Recap: Spillover Networks

- ▣ The generalized variance decomposition of a VAR gives a directed, weighted network that does not depend on the ordering
- ▣ Two thirds of bank volatility risk comes from other banks; US banks transmit, HSBC and Romanian banks receive
- ▣ Connectedness jumps in crises and decays slowly: a monitoring tool for supervisors
- ▣ Choices matter: VAR order, horizon, window length and volatility versus returns (Seminar 18, B5)

The Idea of the Financial Risk Meter

- For each firm j , a quantile regression at $\tau = 5\%$ of its return on the returns of **all** other firms and on lagged macro variables [Mihoci, Althof, Chen & Härdle \(2020\)](#)
 - ▶ With 100 firms and a 63-day window there are more regressors than days: a penalty is needed
- **LASSO** (Least Absolute Shrinkage and Selection Operator) quantile regression [Belloni & Chernozhukov \(2011\)](#)
 - ▶ $\hat{\beta}_j(\lambda) = \arg \min_{\beta} \frac{1}{n} \sum_t \rho_{\tau} \left(X_{j,t} - \beta \mathbf{o} - \beta^{\top} Z_{-j,t} \right) + \lambda \|\beta\|_1$
 - ▶ λ switches off weak links: a larger λ leaves fewer active coefficients
- The **FRM** (Financial Risk Meter) is the average of the selected λ_j over firms, designed as an indicator of tail interdependence and systemic stress
 - ▶ Built for US financial firms [Mihoci, Althof, Chen & Härdle \(2020\)](#), emerging markets [Ben Amor, Althof & Härdle \(2022\)](#) and China [Wang, Althof & Härdle \(2023\)](#)

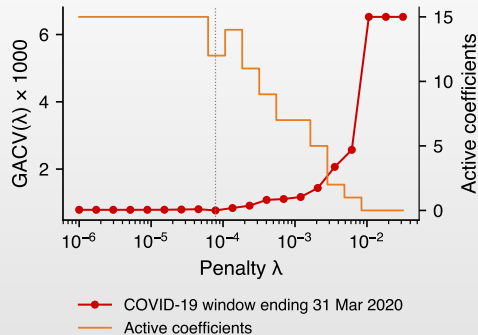
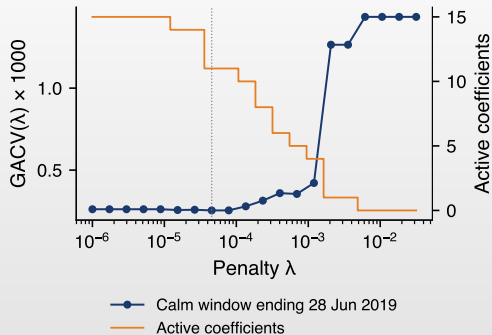
Choosing the Penalty: GACV

- **GACV** (Generalized Approximate Cross-Validation) trades fit against the number of active coefficients

►
$$\text{GACV}(\lambda) = \frac{\sum_t \rho_\tau \left(X_{j,t} - \hat{X}_{j,t}(\lambda) \right)}{n - \text{df}(\lambda)}, \text{ df} = \text{number of nonzero coefficients}$$

- $\hat{\lambda}_j = \arg \min_{\lambda} \text{GACV}(\lambda)$ on a grid; $\text{FRM}_t = \frac{1}{J} \sum_j \hat{\lambda}_{j,t}$
- Our version: 13 banks, macro variables S&P 500, Euro Stoxx 50 and VIX at $t - 1$, 63-day windows every 10 days
 - 15 regressors for 63 days: far fewer than in the published FRM
- Extensions: FRM based on expectiles [Ren, Lu, Li & Härdle \(2022\)](#), for cryptocurrencies [Ren, Althof & Härdle \(2022\)](#) and for the EU energy sector [Toma, Dima, Grecu, Pele & Lessmann \(2026\)](#)

GACV in a Calm and in a Crisis Window

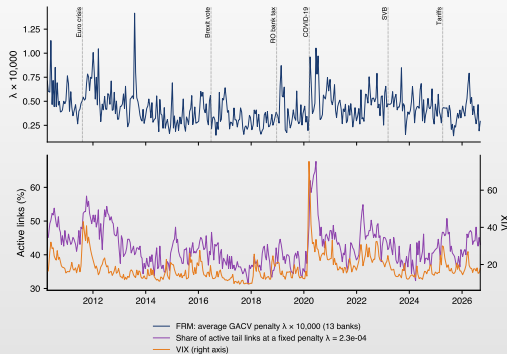


- JPMorgan regressed on the other 12 banks and three lagged macro variables, 63 days; dotted line: selected λ; orange: active coefficients

Reading the GACV Curves

- Calm window (to 28 June 2019): $\hat{\lambda} = 4.5 \times 10^{-5}$, 11 of 15 coefficients active
 - ▶ COVID-19 window (to 31 March 2020): $\hat{\lambda} = 7.8 \times 10^{-5}$, 12 active
- The curve is flat for small λ and rises steeply once key links are switched off
- In the crisis the whole curve is about three times higher: the check loss is in return units, and returns are larger
- With few regressors, the minimum lies where almost no penalty is applied: the selection is noisy

The Financial Risk Meter of Thirteen Banks



- Top: average GACV penalty over 13 banks; bottom: share of active tail links when λ is held fixed, with the VIX; 390 windows of 63 days

Reading Our Financial Risk Meter

- The average GACV penalty is noisy; its correlation with the VIX is only 0.35
 - ▶ With 15 regressors and 63 days the fit hardly needs a penalty; the published FRM uses about 100 firms, so that the penalty must do real work
- At a fixed penalty $\lambda = 2.3 \times 10^{-4}$, the share of active tail links is a clearer signal: correlation 0.62 with the VIX
 - ▶ From 31% (2017-11-21) to 68% (2020-06-17); average 42%
- Lesson: a tail network needs many nodes; with a small panel, count the links rather than average the penalties

Recap: the Financial Risk Meter

- Each firm's 5% quantile is regressed on all other firms with an L1 penalty; the penalty λ chosen by GACV summarises tail interdependence
- The FRM is designed to signal systemic stress; it relies on large cross-sections of firms
- On 13 banks the penalty is noisy, while the number of active tail links follows the VIX
- Advanced material: Quantinar course on the FRM (end of chapter)

Climate Risk as a Systemic Risk

- ▣ **Physical risk:** losses from floods, droughts, heat and storms that damage collateral and borrowers
 - ▶ **Transition risk:** losses from the move to a low-carbon economy (carbon prices, regulation, technology)
- ▣ “Green swan” risks: potentially extremely disruptive events that could be behind the next systemic financial crisis; backward-looking risk models cannot anticipate them [Bolton et al. \(2020\)](#)
- ▣ A network climate stress test maps investors’ exposures to climate-policy-relevant sectors and adds second-round effects through the financial network [Battiston et al. \(2017\)](#)
- ▣ Hence scenario analysis instead of historical estimation: the past sample contains no energy transition

The ECB Climate Risk Stress Test of 2022



Seat of the European Central Bank, Frankfurt, 2015

Photo: DXR (2015); CC BY-SA 4.0; Wikimedia Commons

- 104 significant banks took part; a bottom-up projection of losses covered 41 of them [ECB Banking Supervision \(2022\)](#)
- About 60% of banks had no climate risk stress-testing framework; only 20% considered climate risk when granting loans
- Almost two thirds of banks' income from non-financial corporate customers came from greenhouse gas-intensive industries
- Credit and market losses of around EUR 70 billion for the 41 banks, under a short-term disorderly transition and two physical risk scenarios
 - ▶ The ECB states that this significantly understates the risk: data gaps, rudimentary models, no second-round effects

What Is a Stress Test?



The Eccles Building of the Federal Reserve Board,
Washington, D.C.

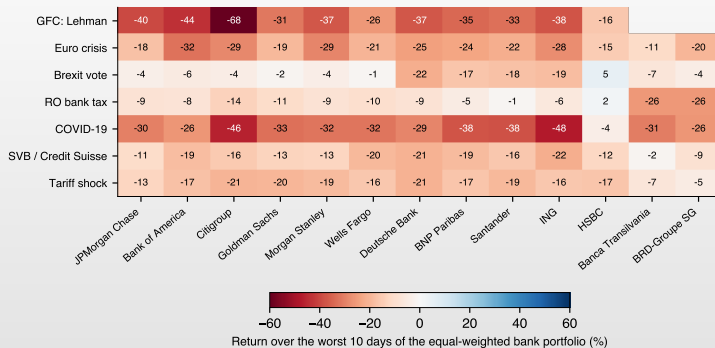
Photo: Federal Reserve (2011); public domain; Wikimedia
[Commons](#)

- A stress test asks: how much would we lose in a **severe but plausible** scenario?
 - ▶ Supervisory tests of large banks: [Federal Reserve stress tests](#) in the US, EU-wide tests of the [European Banking Authority](#)
- Three designs
 - ▶ **Historical**: replay a past crisis on today's portfolio
 - ▶ **Hypothetical**: shock chosen risk factors and propagate through a model
 - ▶ **Reverse**: fix the loss that would break us and find the most plausible scenario that produces it
- Global systemically important banks also hold extra capital, set by the [Basel Committee \(2018\)](#) methodology

Historical Scenarios: Method

- ▣ Seven episodes, each a window of a few weeks around a known shock
 - ▶ Lehman (autumn 2008), the euro crisis (summer 2011), the Brexit vote (June 2016), the Romanian bank tax (December 2018), COVID-19 (March 2020), SVB and Credit Suisse (March 2023), the tariff shock (April 2025)
- ▣ Inside each window we find the **worst 10 trading days** of the equal-weighted bank portfolio and report every bank's return over those days
- ▣ The 2008 episode uses the 11 international banks only: the Romanian series start in 2010
- ▣ Ten days match the horizon of market-risk capital (Chapter 7)

Seven Historical Stress Scenarios



Return (%) of each bank over the worst 10 days of the equal-weighted bank portfolio in each episode; blank: no data

Reading the Historical Scenarios

- Portfolio loss over the worst 10 days: Lehman 38.3% (2008-11-10 to 2008-11-21), COVID-19 32.5%, euro crisis 22.7%
 - ▶ Tariffs 16.1%, SVB 15.0%, Romanian bank tax 10.6%, Brexit 8.2%
- Worst single cells: Citigroup −68% in 2008, ING −48% in March 2020
- Shocks are local as well as global: Brexit hit mainly European banks; the Romanian bank tax hit Romanian banks hardest (−26% and −26%), during a December 2018 sell-off that also cost US banks −8% to −14%
- A stress test built only on global crises would miss the domestic political shock

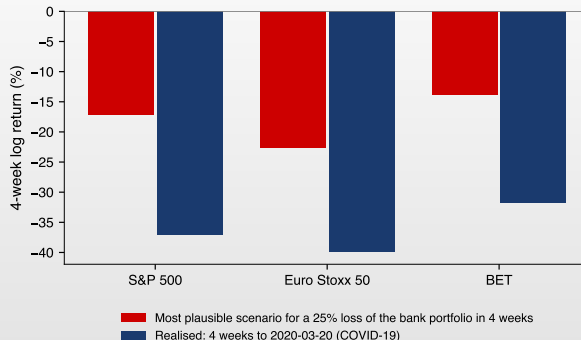
Reverse Stress Testing

- Factor model: weekly bank returns $X_i = \beta_i^\top f + e_i$, factors $f = (\text{S\&P 500, Euro Stoxx 50, BET})$, over h weeks
 $f \sim N(0, h\Sigma)$
 - ▶ Loss of the equal-weighted portfolio: $L = -b^\top f$, with b the average of the β_i (idiosyncratic terms diversify away)
- The most plausible scenario with loss ℓ^* minimises the Mahalanobis distance $f^\top (h\Sigma)^{-1} f$ subject to $-b^\top f = \ell^*$
 - ▶ $f^* = -\ell^* \frac{h\Sigma b}{b^\top h\Sigma b}$, $d^* = \frac{\ell^*}{\sqrt{b^\top h\Sigma b}}$
- d^* counts portfolio standard deviations: the larger it is, the less plausible the loss under the model

Worked Example: Reverse Stress Test with Two Factors

- Betas $b = (1.2, 0.8)$; four-week covariance of the factors $h\Sigma = \begin{pmatrix} 16 & 6 \\ 6 & 9 \end{pmatrix}$ (%²); target loss $\ell^* = 20\%$
 - ▶ $h\Sigma b = (1.2 \cdot 16 + 0.8 \cdot 6, 1.2 \cdot 6 + 0.8 \cdot 9) = (24, 14.4)$
 - ▶ $b^\top h\Sigma b = 1.2 \cdot 24 + 0.8 \cdot 14.4 = 40.32$, standard deviation 6.35%
- Most plausible scenario and its distance
 - ▶ $f^* = -20 \cdot (24, 14.4)/40.32 = (-11.9\%, -7.1\%)$, $d^* = 20/6.35 = 3.15$
 - ▶ Normal probability of a loss at least this large: $\Phi(-3.15) = 0.08\%$ per four weeks
- The scenario loads more on the factor with the larger beta and variance, and both factors fall together because they are correlated

The Most Plausible Scenario for a 25% Loss



- Weekly data 2010–2026 (871 weeks), four-week horizon; red: f^* for $\ell^* = 25\%$; blue: factor moves in the worst four weeks of 2020

Reading the Reverse Stress Test

- Portfolio betas: $b = (0.45, 0.65, 0.18)$; four-week standard deviation 5.79%
 - ▶ A 25% loss needs S&P 500 -17.2% , Euro Stoxx 50 -22.6% , BET -13.9% together
- Distance $d^* = 4.31$: under the Normal model its probability is 8.0×10^{-6} per four weeks, practically impossible
- Yet in the four weeks to 2020-03-20 the portfolio lost 52.0% in log terms, with factor falls larger than f^* (the model implies 48.5%)
- The shape of the scenario is informative; its probability is not: heavy tails and rising correlations make Normal probabilities far too small

Limits of Stress Tests

- ▣ Scenarios are chosen by people: the crisis that comes is rarely the one that was tested
- ▣ Second-round effects (fire sales, funding runs, counterparty losses) are hard to model and usually left out, as the ECB noted for its climate test
- ▣ Market-based measures (MES, CoVaR, connectedness) react quickly but only after prices move; balance-sheet stress tests look forward but slowly
- ▣ Early-warning research combines both, e.g. entropy-based measures of market risk [Pele, Lazar & Dufour \(2017\)](#)

Recap: Stress Testing

- ▣ Historical, hypothetical and reverse stress tests answer different questions
- ▣ Across seven episodes, the worst 10-day loss of a bank portfolio ranged from 8.2% to 38.3%
- ▣ Reverse stress tests give the most plausible direction of a disaster, not its probability
- ▣ Climate scenarios extend stress testing to risks that have no history

An Open Question

Does network connectedness measure systemic risk, or only volatility?

- Connectedness indices are monitored as early-warning tools, yet the index is built from a VAR on volatilities and rises whenever volatility rises
 - ▶ Live research: global bank networks from regularised VARs [Demirer et al. \(2018\)](#); connectedness in the tails differs from connectedness in the mean [Ando, Greenwood-Nimmo & Shin \(2022\)](#)
- Formal version, for future bank stress y_{t+h} , total connectedness C_t and current volatility σ_t :
$$y_{t+h} = a + \beta_C C_t + \gamma \ln \sigma_t + e_{t+h}, \quad H_0: \beta_C = 0, \quad H_1: \beta_C > 0$$
 - ▶ Falsified if C_t loses its predictive power once current volatility is controlled
- Identification: a network edge is a share of forecast error variance, not a causal exposure
 - ▶ A common shock raises volatility and connectedness together, with no bank infecting another

The Discovery Loop, Step by Step (1/2)

- **Literature:** map what is known, then verify every reference
 - ▶ “Which papers test whether connectedness indices predict bank distress out of sample, beyond volatility? Give DOI, sample, horizon and controls.”
 - ▶ Resolve each DOI on [Crossref](#) and check that the paper really controls for volatility; search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses:** competing readings of a rise in connectedness, each with its own test
 - ▶ “Give competing interpretations of a rise in total connectedness (common shock, contagion, artefact of rolling windows) and a test that discriminates between them.”
 - ▶ E.g. a common shock fades with volatility; contagion has a direction: shocks of net transmitters precede those of receivers
- **Data and code:** on the course data, replicating a known number first
 - ▶ “Fit a VAR(2) to the log Parkinson volatilities of the 13 banks in data/market on 250-day windows and compute the generalized total connectedness with $H = 10$.”
 - ▶ Replicate first: mean 62.0% and maximum 79.3% (2020-08-27), as in this chapter

The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed before the predictive regressions are run
 - ▶ “Pre-register: horizons 21/63/126 days, volatility and VIX controls, windows 150/250/500 days, HAC lags at least the overlap, out-of-sample evaluation from 2018.”
 - ▶ Every cell of the grid is reported; the family is corrected with Holm or controlled with the FDR
- **Critique**: ask the assistant to attack the result
 - ▶ “Act as a referee: is the predictive power of connectedness just the persistence of volatility? Which regression would show it?”
 - ▶ Expect: add current volatility as a control, then test the increment of R^2 and of the out-of-sample loss
- **Report**: standardised effects with HAC t -statistics, every cell of the grid, what failed
 - ▶ The prompts that shaped the design go into AI_USE.md

What the Human Must Check

- **Non-stationary inputs:** a VAR on price levels, or on trending series, produces spurious connectedness
 - ▶ Test: unit-root tests on every input; all roots of the fitted VAR inside the unit circle in every window
- **Ordering:** Cholesky decompositions change with the order of the banks
 - ▶ Test: permute the columns; the generalized table must not change
- **Look-ahead:** a window labelled by its first date, or a lag order chosen on the full sample
 - ▶ Test: the index at t must not move when data after t are deleted
- **Overlap:** rolling indices and forward outcomes overlap, so OLS t -statistics are inflated
 - ▶ Test: Newey–West lags at least equal to the overlap, or a block bootstrap
- **Dimension:** 13 banks and $p = 2$ give 27 coefficients per equation from 250 days; more banks need shrinkage [Demirer et al. \(2018\)](#)
 - ▶ Test: compare with a LASSO VAR; unstable edges signal estimation noise, not structure

Mini-Case: Is Connectedness More Than Volatility?

- A typical assistant proposal: "high total connectedness warns of future bank stress"; we test it on 731 weekly dates, 2011–2026
 - ▶ y_{t+h} : log mean Parkinson volatility of the 13 banks over the next $h = 63$ days; control: the same over the last 20 days; standardised variables, Newey–West with 13 lags
- Connectedness alone: $\hat{\beta}_C = 0.40$ ($t = 4.58$), $R^2 = 0.16$: the proposal looks right
- With current volatility: $\hat{\beta}_C = 0.08$ ($t = 0.72$), $\hat{\gamma} = 0.53$ ($t = 4.13$); $R^2 = 0.343$ vs 0.339 with volatility alone
 - ▶ Correlation of C_t with current volatility: 0.59; for the next 63-day maximum drawdown of the bank portfolio, $\hat{\beta}_C = -0.09$ ($t = -1.27$) even alone
- Interpretation: in our sample the predictive content of the index is the persistence of volatility; network information may still matter in the tails [Ando, Greenwood-Nimmo & Shin \(2022\)](#) or for the direction of transmission

Your Turn: a Project Seed

- **Question:** does **tail** connectedness predict bank distress beyond volatility, out of sample?
- **Replicate first:** our rolling index (mean 62.0%, spring 2020 78.4%), then one headline connectedness figure of [Demirer et al. \(2018\)](#) on their sample
 - ▶ Extend: quantile connectedness at $\tau = 5\%$ [Ando, Greenwood-Nimmo & Shin \(2022\)](#), out-of-sample evaluation from 2018, MES and SRISK as alternative targets
- **Deliverables:** a pre-registered grid committed before the regressions, a notebook that regenerates every number, HAC inference and out-of-sample losses for every cell
- **AI log:** prompts that shaped the design in AI_USE.md; each wrong claim (a VAR on price levels, OLS t -statistics on overlapping windows) and the test that caught it in AI_ERRORS.md ([templates](#))

Which Bank Is Most Systemic?

Bank	VaR 1%	MES 5%	$\Delta\text{CoVaR 1\%}$	Net spillover	L^*
JPMorgan	4.83	3.22	4.62	12.6	7.0
Bank of America	5.99	3.85	4.60	12.4	6.2
Citigroup	5.87	4.03	3.81	15.6	6.0
Goldman Sachs	4.81	3.34	4.12	2.5	6.8
Morgan Stanley	5.83	4.04	4.00	7.9	6.0
Wells Fargo	5.78	3.32	4.89	2.1	6.8
Deutsche Bank	7.19	4.54	4.24	-11.3	6.1
BNP Paribas	6.18	4.53	4.84	6.2	6.1
Santander	5.83	4.23	4.42	-4.0	6.4
ING	7.16	4.81	4.42	7.5	5.8
HSBC	4.66	2.54	4.07	-25.8	8.3
Banca Transilvania	5.06	3.14	—	-12.0	6.5
BRD-Groupe SG	4.42	2.69	—	-13.8	7.1

□ Daily losses in %; net spillover of volatility (to minus from); L^* : break-even leverage with $k = 8\%$

Reading the Comparison

- ▣ Rank correlations across the eleven international banks
 - ▶ MES and ΔCoVaR : -0.01 ; MES and net spillover: -0.01 ; ΔCoVaR and net spillover: -0.03
 - ▶ VaR and MES: 0.88
- ▣ The measures answer different questions: exposure to the market, contribution to the system, transmission of volatility
- ▣ No single ranking: supervisors combine several indicators with size and balance-sheet data, as in the [Basel Committee \(2018\)](#) methodology
- ▣ Romanian banks: moderate MES, net receivers of volatility, break-even leverage close to that of US banks

Key Takeaways

- ▣ Systemic risk lives in the joint tail: exposure (MES, SRISK), contribution (ΔCoVaR) and connectedness (spillover and tail networks)
- ▣ European banks have the largest MES in our sample; US banks are the main net transmitters of volatility
- ▣ VaR and ΔCoVaR rank banks differently; the rankings are uncertain, so report intervals
- ▣ Market-based measures jump in crises; stress tests and balance sheets are needed to see risk building up
- ▣ Romanian banks are mostly driven by domestic shocks and receive more than they transmit

Common Pitfalls

- ▣ Reading ΔCoVaR as causal: a common shock creates the same pattern
- ▣ Ranking banks by point estimates when the bootstrap intervals overlap
- ▣ Including the bank itself in the system portfolio: its own return inflates the link mechanically
- ▣ Orthogonal variance decompositions whose results depend on the order of the banks
- ▣ Reporting SRISK without balance-sheet data, or reverse-stress probabilities from a Normal model

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Financial Risk Meter for Emerging Markets
- ▣  Dynamic Crypto Networks
- ▣  Network Centrality

Code and data

- ▣ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_18

References (1/4)

- ▣ Acharya, V.V., Engle, R., Richardson, M. (2012). Capital shortfall: a new approach to ranking and regulating systemic risks. *American Economic Review*, 102(3), 59–64.
- ▣ Acharya, V.V., Pedersen, L.H., Philippon, T., Richardson, M. (2017). Measuring systemic risk. *Review of Financial Studies*, 30(1), 2–47.
- ▣ Adrian, T., Brunnermeier, M.K. (2016). CoVaR. *American Economic Review*, 106(7), 1705–1741.
- ▣ Ando, T., Greenwood-Nimmo, M., Shin, Y. (2022). Quantile connectedness: modeling tail behavior in the topology of financial networks. *Management Science*, 68(4), 2401–2431.
- ▣ Basel Committee on Banking Supervision (2018). Global systemically important banks: revised assessment methodology and the higher loss absorbency requirement. Bank for International Settlements.
- ▣ Battiston, S., Mandel, A., Monasterolo, I., Schütze, F., Visentin, G. (2017). A climate stress-test of the financial system. *Nature Climate Change*, 7(4), 283–288.
- ▣ Belloni, A., Chernozhukov, V. (2011). ℓ_1 -penalized quantile regression in high-dimensional sparse models. *Annals of Statistics*, 39(1), 82–130.
- ▣ Ben Amor, S., Althof, M., Härdle, W.K. (2022). Financial Risk Meter for emerging markets. *Research in International Business and Finance*, 60, 101594.
- ▣ Billio, M., Getmansky, M., Lo, A.W., Pelizzon, L. (2012). Econometric measures of connectedness and systemic risk in the finance and insurance sectors. *Journal of Financial Economics*, 104(3), 535–559.

References (2/4)

- ▣ Bolton, P., Després, M., Pereira da Silva, L.A., Samama, F., Svartzman, R. (2020). [The green swan: central banking and financial stability in the age of climate change](#). Bank for International Settlements.
- ▣ Brownlees, C., Engle, R.F. (2017). [SRISK: a conditional capital shortfall measure of systemic risk](#). *Review of Financial Studies*, 30(1), 48–79.
- ▣ Demirer, M., Diebold, F.X., Liu, L., Yilmaz, K. (2018). [Estimating global bank network connectedness](#). *Journal of Applied Econometrics*, 33(1), 1–15.
- ▣ Diebold, F.X., Yilmaz, K. (2009). [Measuring financial asset return and volatility spillovers, with application to global equity markets](#). *Economic Journal*, 119(534), 158–171.
- ▣ Diebold, F.X., Yilmaz, K. (2012). [Better to give than to receive: predictive directional measurement of volatility spillovers](#). *International Journal of Forecasting*, 28(1), 57–66.
- ▣ Diebold, F.X., Yilmaz, K. (2014). [On the network topology of variance decompositions: measuring the connectedness of financial firms](#). *Journal of Econometrics*, 182(1), 119–134.
- ▣ ECB Banking Supervision (2022). [Banks must sharpen their focus on climate risk, ECB supervisory stress test shows](#). Press release, 8 July 2022.
- ▣ FDIC (2023). [FDIC creates a Deposit Insurance National Bank of Santa Clara to protect insured depositors of Silicon Valley Bank, Santa Clara, California](#). Press release, 10 March 2023.
- ▣ Federal Reserve Board (2023). [Federal Reserve Board announces it will make available additional funding to eligible depository institutions](#). Press release, 12 March 2023.

References (3/4)

- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). *Applied Quantitative Finance*, 3rd ed. Springer.
- ▣ Härdle, W.K., Wang, W., Yu, L. (2016). *TENET: Tail-Event driven NETwork risk*. *Journal of Econometrics*, 192(2), 499–513.
- ▣ Koenker, R., Bassett, G. (1978). *Regression quantiles*. *Econometrica*, 46(1), 33–50.
- ▣ Koop, G., Pesaran, M.H., Potter, S.M. (1996). *Impulse response analysis in nonlinear multivariate models*. *Journal of Econometrics*, 74(1), 119–147.
- ▣ Mihoci, A., Althof, M., Chen, C.Y.-H., Härdle, W.K. (2020). *FRM Financial Risk Meter*. In *The Econometrics of Networks*, Advances in Econometrics 42, Emerald, 335–368.
- ▣ Parkinson, M. (1980). *The extreme value method for estimating the variance of the rate of return*. *Journal of Business*, 53(1), 61–65.
- ▣ Pele, D.T., Lazar, E., Dufour, A. (2017). *Information entropy and measures of market risk*. *Entropy*, 19(5), 226.
- ▣ Pesaran, H.H., Shin, Y. (1998). *Generalized impulse response analysis in linear multivariate models*. *Economics Letters*, 58(1), 17–29.

References (4/4)

- ▣ Ren, R., Althof, M., Härdle, W.K. (2022). [Financial Risk Meter for cryptocurrencies and tail risk network-based portfolio construction](#). *Singapore Economic Review*, 70(4), 901–927.
- ▣ Ren, R., Lu, M., Li, Y., Härdle, W.K. (2022). [Financial Risk Meter FRM based on expectiles](#). *Journal of Multivariate Analysis*, 189, 104881.
- ▣ Toma, F.-M., Dima, D.M., Grecu, R., Pele, D.T., Lessmann, S. (2026). [Energy Financial Risk Meter: systemic risk in the EU energy sector](#). SSRN working paper.
- ▣ Wang, R., Althof, M., Härdle, W.K. (2023). [A financial risk meter for China](#). *Emerging Markets Review*, 56, 101052.